Institute of technology of Cambodia

Department GIC

Course Enrollment Project

I4-GIC-Group C

Individual Report

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1. Project Overview

My individual project's designed and developed a University Course Enrollment and Classroom Scheduling System.
The system helps manage students, lecturers, courses, enrollment, attendance, scheduling, and reporting in a structured and secure way.

I implemented the project using Spring Boot and followed a service-based layered architecture, separating business logic into clear service modules.

2. Project Objectives

The main objectives of my project are:

- To manage users (Admin, Lecturer, Student) securely
- To allow students to enroll in courses
- To manage course and classroom scheduling
- To detect scheduling conflicts automatically
- To track student attendance by session
- To record audit logs for system activities
- To export data to Google Sheets
- To manage file uploads and storage

3. System Architecture

I used a Layered Architecture consisting of:

- Controller Layer (API endpoints)
- Service Layer (business logic)
- Repository Layer (database access)

Each service is implemented using:

- Service.java → interface
- ServiceImpl.java → implementation

This approach improves code organization, readability, and maintainability.

4. Service Modules Description

4.1 Attendance Service

I created this module to manage student attendance.

Files:

- AttendanceService.java
- AttendanceServiceImpl.java

Functions:

- Mark student attendance
- Retrieve attendance records
- Generate attendance summaries

4.2 Audit Service

This module is used to track system activities.

Files:

- AuditService.java
- AuditServiceImpl.java

Functions:

- Log user actions (login, update, delete)
- Improve system accountability and monitoring

4.3 Authentication & User Management

I implemented authentication and role-based access control.

Files:

- AuthenticationService
- AdminService
- LecturerService
- StudentService
- UserService

Functions:

- User login and authentication
- Role-based authorization
- Manage admin, lecturer, and student accounts

4.4 Course & Enrollment Service

This module manages courses and student enrollment.

Files:

- CourseService
- EnrollmentService

Functions:

- Create and update courses
- Enroll students in courses
- Validate enrollment rules

4.5 Google Sheets Integration

I integrated Google Sheets for online reporting.

Files:

- GoogleSheetsService
- GoogleSheetsServiceImpl

Functions:

- Export attendance and enrollment data
- Generate reports for administrators

4.6 Scheduling Service

This module handles classroom scheduling.

Files:

- ClassroomService
- ScheduleService
- ConflictDetectionService

Functions:

- Assign classrooms
- Create schedules
- Detect time and room conflicts automatically

4.7 Session Management

This module manages class sessions.

Files:

- SessionService
- SessionServiceImpl

Functions:

- Create academic sessions
- Link sessions with schedules and attendance

5. Technologies Used

- Java
- Spring Boot
- Spring Security
- JPA / Hibernate
- MySQL (or PostgreSQL)
- Google Sheets API
- RESTful APIs

6. What I Learned From This Project

- Designing a clean service-based architecture
- Implementing role-based authentication
- Managing complex relationships (courses, schedules, sessions)
- Detecting logical conflicts in scheduling
- Writing maintainable and scalable backend code

7. Conclusion

This individual project allowed me to apply real-world backend development concepts using Spring Boot.
By separating responsibilities into well-defined services, I built a system that is secure, scalable, and easy to maintain, reflecting practical university management workflows.